

DBMS Module 1 – Easy Notes

1. Introduction to Database

What is Data?

Data means raw facts or values. Examples: numbers, names, marks, phone numbers.

Example:

- 10
- ABC
- 95

By itself, data may not give clear meaning.

What is a Record?

A record is a collection of related data.

Example: | Name | Phone | City | |---|---|---| | Rahul | 9876543210 | Bangalore |

Here, all values together give meaningful information.

What is a Database?

A database is a collection of related data stored in an organized way.

Examples:

- Student database
- Bank database
- Hospital database
- Railway reservation database

A database:

- stores meaningful data
 - is designed for a specific purpose
 - can be small or very large
-

Terms Used in Tables

Term	Meaning
Row	Record / Tuple
Column	Field / Attribute
Table	Relation

2. DBMS (Database Management System)

What is DBMS?

DBMS is software used to:

- create databases
- store data
- update data
- retrieve data
- manage databases

Examples:

- MySQL
 - Oracle
 - SQL Server
 - PostgreSQL
-

Functions of DBMS

1. Defining Database

Specifying:

- data types
- structure
- constraints

2. Constructing Database

Storing data into storage devices.

3. Manipulating Database

Operations like:

- insert
- update
- delete
- search
- generate reports

4. Sharing Database

Allows many users to use the same database together.

Database System

Database + DBMS software together are called a Database System.

3. File System vs Database System

File Processing System Problems

In old systems:

- each department stored separate files
- same data repeated many times
- difficult to maintain

Problems:

- data duplication
 - inconsistency
 - difficult searching
 - poor security
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Database Approach

In DBMS:

- data is stored centrally
 - multiple users can access same data
 - redundancy is reduced
 - data becomes consistent
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4. Characteristics of DBMS

1. Self-Describing Nature

DBMS stores:

- data
- structure of data
- rules and constraints

This information is called Metadata.

Metadata = Data about data.

2. Program-Data Independence

Changing database structure usually does not require changing all programs.

Advantage:

- easier maintenance
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3. Data Abstraction

Users do not need to know internal storage details.

They only see required information.

4. Multiple Views

Different users can see different parts of the database.

Example:

- Student sees marks
 - Teacher sees attendance and marks
 - Admin sees complete details
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5. Multiuser Access

Many users can access database at the same time.

DBMS controls access properly.

Example: Two people should not book the same seat.

Important Transaction Properties

Atomicity

Either all operations happen or none happen.

Isolation

Transactions work independently without affecting each other.

5. Users of DBMS

1. Database Administrator (DBA)

Responsible for:

- security
 - backup
 - permissions
 - performance
-

2. Database Designers

Design database structure.

3. End Users

People who use the database.

Types of End Users

Casual Users

Use database occasionally. Example: managers.

Naive Users

Use predefined applications. Example:

- bank tellers
- railway booking staff

Sophisticated Users

Experts like:

- engineers
- analysts
- researchers

Standalone Users

Maintain personal databases.

4. System Analysts and Programmers

- analyze requirements
 - develop software applications
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6. Advantages of DBMS

1. Controls Data Redundancy

Avoids repeated data.

2. Better Security

Unauthorized users can be restricted.

3. Backup and Recovery

Data can be recovered after failure.

4. Multiple User Interfaces

Different interfaces for different users.

5. Integrity Constraints

Ensures correct and valid data.

Example: Age cannot be negative.

6. Efficient Data Access

Searching and updating become faster.

7. Represents Relationships

DBMS can manage relationships between data easily.

7. Disadvantages of DBMS

- High cost
 - Requires hardware and software
 - Complex system
 - Needs trained people
 - Overhead for security and recovery
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8. Data Models

What is a Data Model?

A data model defines:

- structure of data
 - relationships
 - constraints
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Types of Data Models

1. High-Level / Conceptual Model

Easy for users to understand.

Uses:

- entities
- attributes
- relationships

Example: ER Model

2. Physical Data Model

Describes how data is stored physically.

Focuses on:

- files

- storage
 - indexing
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3. Representational / Relational Model

Uses tables to represent data.

Most commonly used model.

9. Schema and Instance

Schema

Structure/design of database.

Example: STUDENT(RollNo, Name, Age)

Schema changes very rarely.

Instance (Snapshot)

Actual data present at a particular time.

Example: | RollNo | Name | |---|---| | 1 | Ravi |

Data can change frequently.

Schema Evolution

Changing database schema.

Example: Adding a new column: Date_of_Birth

10. Three Schema Architecture

Purpose: Separate users from physical storage.

1. Internal Level

Lowest level.

Describes:

- physical storage
 - files
 - indexes
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2. Conceptual Level

Middle level.

Describes:

- whole database structure
 - entities
 - relationships
 - constraints
-

3. External/View Level

Highest level.

Different users see different views.

11. Data Independence

Logical Data Independence

Changes in conceptual schema should not affect user views.

Example: Adding new attribute.

Physical Data Independence

Changes in physical storage should not affect conceptual schema.

Example: Changing storage method.

12. Database Languages

1. DDL (Data Definition Language)

Used to define database structure.

Commands:

- CREATE
 - ALTER
 - DROP
-

2. DML (Data Manipulation Language)

Used to manipulate data.

Commands:

- INSERT
 - UPDATE
 - DELETE
 - SELECT
-

3. SDL (Storage Definition Language)

Defines internal storage.

4. VDL (View Definition Language)

Defines user views.

SQL

Most DBMS use SQL.

SQL = Structured Query Language.

13. Types of DML

High-Level DML

Easy to use. Works on multiple records together.

Example: SQL queries.

Low-Level DML

Works record by record. Usually embedded in programming languages.

14. DBMS Interfaces

Menu-Based Interface

Uses menus.

Forms-Based Interface

Users fill forms.

GUI (Graphical User Interface)

Uses graphics and diagrams.

Natural Language Interface

Users type commands in normal language.

Speech Interface

Uses voice input/output.

15. DBMS Components

Query Processor

Processes SQL queries.

Query Optimizer

Chooses best execution method.

Storage Manager

Controls stored data.

Buffer Manager

Manages memory buffers.

Backup & Recovery Manager

Handles recovery after failure.

16. Database Utilities

Loading

Loads existing data into database.

Backup

Creates backup copies.

Reorganization

Improves storage performance.

Performance Monitoring

Checks database performance.

17. ER Model (Entity Relationship Model)

ER Model is used for database design.

Main concepts:

- Entity
- Attribute
- Relationship

18. Entity

An entity is a real-world object.

Examples:

- Student
- Employee
- Car
- Course

19. Attributes

Attributes describe entities.

Example: EMPLOYEE:

- Name
- Age
- Salary

20. Types of Attributes

1. Simple Attribute

Cannot be divided.

Example: Age

2. Composite Attribute

Can be divided into smaller parts.

Example: Address → Street, City, State

3. Single-Valued Attribute

Only one value.

Example: SSN

4. Multi-Valued Attribute

Multiple values possible.

Example: Phone Numbers

5. Stored Attribute

Stored directly.

Example: Birth Date

6. Derived Attribute

Calculated from other attributes.

Example: Age derived from Birth Date.

7. NULL Value

Value is missing or not applicable.

21. Entity Type and Entity Set

Entity Type

Definition of entities.

Example: EMPLOYEE

Entity Set

Collection of all entities.

Example: All employees in company.

22. Key Attribute

Attribute used to uniquely identify entities.

Example: Employee_ID

Properties:

- unique
 - no duplicates
-

23. Weak Entity

Weak entity does not have its own key.

It depends on another entity.

Example: DEPENDENT depends on EMPLOYEE.

24. Relationship

Relationship connects entities.

Example: EMPLOYEE works for DEPARTMENT.

25. Degree of Relationship

Binary Relationship

Between 2 entities.

Example: EMPLOYEE — DEPARTMENT

Ternary Relationship

Between 3 entities.

Example: SUPPLIER — PART — PROJECT

26. Recursive Relationship

Entity related to itself.

Example: EMPLOYEE supervises EMPLOYEE.

27. Cardinality Constraints

Shows maximum relationship participation.

Types

1:1

One to one.

1:N

One to many.

M:N

Many to many.

28. Participation Constraints

Total Participation

Every entity must participate.

Example: Every employee must belong to a department.

Partial Participation

Participation is optional.

Example: Not every employee is a manager.

29. Attributes of Relationships

Relationships can also have attributes.

Example: WORKS_ON relationship may have:

- Hours
-

30. Naming Rules in ER Model

- Use meaningful names
 - Use singular names
 - Entity names in uppercase
 - Attribute names start with capital letters
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31. UML Class Diagram

In UML:

- Entity → Class
- Entity instance → Object

Class contains:

- class name
 - attributes
 - operations
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32. Important Quick Differences

Schema vs Instance

Schema	Instance
Structure	Actual data
Rarely changes	Changes frequently

DDL vs DML

DDL	DML
Defines structure	Manipulates data
CREATE	INSERT
ALTER	UPDATE

DDL	DML
DROP	DELETE

Strong Entity vs Weak Entity

Strong Entity	Weak Entity
Has own key	No own key
Independent	Depends on another entity

Final Quick Revision

Database

Collection of related data.

DBMS

Software to manage databases.

Metadata

Data about data.

Schema

Database structure.

Instance

Current data in database.

Entity

Real-world object.

Attribute

Property of entity.

Relationship

Connection between entities.

Key Attribute

Uniquely identifies entity.

SQL

Language used in DBMS.

Very Important Exam Topics

- DBMS advantages and disadvantages
 - File system vs DBMS
 - Data models
 - Three-schema architecture
 - Data independence
 - DDL and DML
 - ER model
 - Types of attributes
 - Cardinality constraints
 - Weak entity
 - Keys
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Small Memory Tricks

DBMS Functions

D-C-M-S

- Define
- Construct
- Manipulate
- Share

Three Schema Levels

E-C-I

- External
- Conceptual
- Internal

Languages

DDL → Define DML → Manipulate VDL → View SDL → Storage

End of Simplified Module 1 Notes